

Person Re-Identification

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Abstract

This project addresses the challenge of Person Re-Identification (ReID), a critical task for enabling intelligent surveillance, security, and analytics systems by recognizing individuals across diverse camera views and conditions. The study investigates state-of-the-art methods, the impact of pretraining on large-scale human datasets, such as LUPerson, as well as the creation of various models architectures from scratch. Performances of all models were evaluated on standard datasets like Market-1501 and CUHK-03, using metrics such as Rank-1 and mAP.

1 Introduction

Person Re-Identification is challenging due to variations in pose, lighting, occlusions, and camera settings. Our main goal was to explore state-of-the-art model for this task, and also test new architectures by using various techniques that we'll discuss later.

Related Work

We based our project on two main papers: Unsupervised Pre-training for Person Re-identification^[?] and Person re-identification with a locally aware transformer^[?]. The first one describes a novel method to approach vision tasks regarding people: unsupervised pretraining on the proposed dataset *LUPerson*, that should yield better performances. The second one contains results that represent the state-of-the-art for this task.

2 Our Work

Our idea was to follow the top conference paper Unsupervised Pre-training for Person Re-identification^[?] and see if applying it to a good architecture for this task would help increase performances. After this approach we tried to modify the architecture itself, trying to improve it.

2.1 LA Transformers

We started by implementing the Locally Aware Transformers^[?] (or LA Transformers), as described in the paper. The model architecture is a simple ViT that extracts the features and an Average Pooling layer that aggregates them by row,

adding to each one of them the global token **G**. We used a ViT pretrained on ImageNet as the backbone of the model.

Training

The Figure 1 shows the architecture of the model when training: after the Average Pooling layer, each row is passed through a Classifier that will need to predict the identity of the person in the image. Let k be the number of identities in the dataset, then each classifier will predict a probability distribution over k classes (identities). The idea is that the ViT will learn to extract meaningful features for similar identities, so that at test time same person's images will have similar features and thus high similarity scores.

Testing

Testing and Inference are different from training: we don't need the classifiers anymore, so we can remove them and just use the extracted features after the Average Pooling layer as embeddings. These embeddings will be used to determine if two images contain the same person, using a similarity measure (in our case the dot product).

2.2 LUPerson Pretrain

At this point, we had implemented a state-of-the-art architecture, so we tried increasing its performances by pretraining on the the LUPerson^[?] dataset using MoCo-v2^[?]. In practice, we pre-trained the ViT with a projection head on LUPerson and after this, we fine-tuned the whole model on the Market-1501 dataset as described in the previous section.

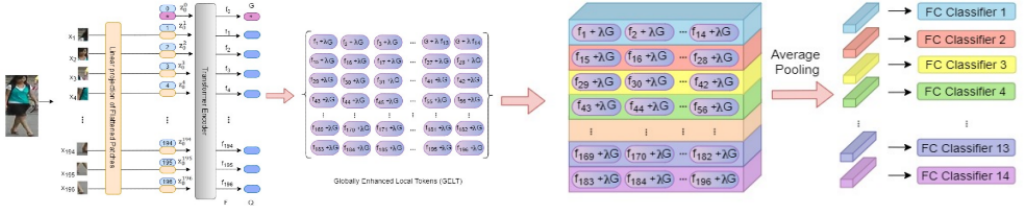
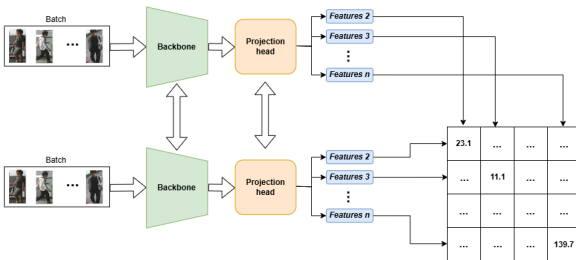


Figure 1: LA Transformers architecture for training.

2.3 Contrastive Learning

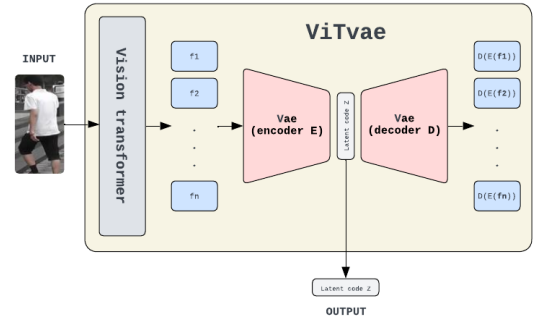
Our third experiment was done by including contrastive learning techniques for the Transformers training, using a Siamese Network (that is, having two models with same architecture and shared parameters). In particular we removed completely the classifiers from the model and we added a projection head with MultiHead Attention, skip connections and a particular type of pooling layer, called Generalized Mean Pooling (GeM)^[7]. This pooling layer showed good performances in other image retrieval tasks, so we decided to try it. The idea is that we can generalize both the formulation for the Average and Max Pooling, by tuning a parameter p that will make the layer behave like one or the other depending on what it is needed. To train, we computed the dot product between embeddings extracted after the GeM pooling layer, resulting in what is called a *Similarity Matrix*. We then used a Clip-like method for training, trying to maximize the *Similarity Matrix* diagonals, which represent the similarity between the same person's images. In order to achieve this we needed to change the dataset, which now had to be composed of pairs of batch of images, where each i -th image in both batches contained the same person from different camera angles. The following Figure shows this architecture.



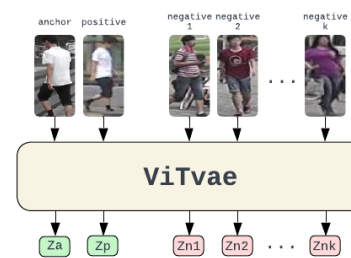
2.4 Custom Architecture

Finally, we tried to create another architecture from scratch. We used a ViT for the features extraction part and then we added a Variational Au-

toencoder (VAE)^[7] to try to create a latent space where the images of the same person are close to each other and the images of different people are distant. We trained the model with a triplet loss function, to induce the anchor image encoding to be close to the positive image encoding and far from the negative images ones. The following Figure shows this architecture.



We used a ViT pretrained on ImageNet and similarly to the LA Transformer we added the global token G to each embedding. The VAE tries to reconstruct the ViT embeddings. It is composed of an Encoder (a series of convolutions) and a Decoder (a series of transposed convolutions). The loss is made of two parts, a standard VAE loss (Reconstruction + KL Divergence) for each element and a *triplet loss*. The triplet loss minimizes the distance between the latent codes of the anchor image and the positive image, and maximizes the distance between the latent codes of the anchor image and all the k negative images.



Given $x_A, x_P, x_{N_1}, x_{N_2}, \dots, x_{N_k}$ the triplet loss is computed as follows:

$$\mathcal{L}_{triplet} = MSE_{loss}(x_A, x_P) - \sum_{i=0}^k (MSE_{loss}(x_A, x_{N_i})).$$

The total loss is the sum of the two losses:

$$\mathcal{L}_{total} = \mathcal{L}_{VAE} + \mathcal{L}_{triplet}$$

where $\mathcal{L}_{VAE}$ is the standard VAE loss.

Unfortunately this model was not able to generalize. We also tried to pretrain it just on the reconstruction loss and then train it on the triplet loss but the results were not good either. We believe that the problem is that the VAE is not able to create a coherent latent space where images of the same person are close to each other and images of different people are distant. This might happen because it has not enough expressive power to learn the complex structure of the ViT embeddings. Since this still needs few adjustments, we decided to leave it as future work.

2.5 Testing measures

In a real world scenario we would have a **query** image, that is an image just captured by a camera and a set of already captured images (**gallery**). We would like to know which identity in the gallery matches the identity of the person in the query. Following this real world scenario, it is very clear that query images should not be in the gallery set, or in other words we should avoid **self-matching**. We noticed a problem with the way testing was performed in the paper^[?]]: they did not avoid self-matching, thus obtaining higher results. Before testing we removed the query images from the gallery set.

3 Results

We tested our model on two of the most famous dataset for this task: market-1501^[?]] and CUHK-03^[?]].

3.1 LA Transformers Tests

In the previous section we talked about the self-matching problem in the testing for LA Transformers. In our implementation we avoided self-matching and found out that the scores claimed in

the paper^[?]] were not accurate: they were on average 24% worse. Also, the mAP was being computed over just the first 10 samples, which is not a standard practice, so we corrected it. These results are shown in the following Table.

Dataset	Model	rank-1	mAP
market-1501	Paper Results	98.27	94.46
	Our Results	89.6	69.2
CUHK-03	Paper Results	98.7	96.4
	Our Results	83.4	48.5

3.2 Other Tests

We observed an increase in performances when pre-training on the LUPerson dataset and a decrease when using contrastive learning techniques. This last model was the hardest to train, and in fact we faced many problems with overfitting. We tried different values for Dropout, Weight Decay and Learning Rate, other than freezing the backbone to train just the projection heads. We also tried to train with all possible pairs of images, rather than having unique classes in each batch, but nothing really helped.

4 Conclusions

This was a very interesting project, which allowed us to explore the ViT architecture and its applications in the field of PersonReID. Even tho we didn't reach the state-of-the-art results, we did learn a lot about the architecture and the field itself. We also proved that the SOTA architecture that was on the PapersWithCode website was not accurate, which we think is an important result.

Future Work

For the future we would like to try and implement other architectures: the current (real) SOTA for this task is still given by models that embed images and then try to measure the similarity between them, but the LA Transformers architecture is perhaps too simplistic.

4.1 Project Roles

The group worked collaboratively on all aspects of the project, but each member had specific tasks where they took the lead. Stefano Saravalle focused on implementing the LA Transformers model, handled the Contrastive Learning task,

Dataset	Model	rank-1	rank-5	rank-10	mAP
market-1501	SOTA	98.0	98.9	-	95.5
	LA Transformers	89.6	97.4	98.1	69.2
	LUPerson Pretrain	89.2	97.8	98.4	70.0
	Contrastive Trans.	49.1	66.9	73.4	10.1
CUHK-03	SOTA	-	-	-	94.3
	LA Transformers	83.4	92.8	96.1	48.5
	LUPerson Pretrain	84.5	95.0	95.0	48.2
	Contrastive Trans.	74.1	87.8	91.7	27.8

Table 1: Test results on market-1501 and CUHK-03 datasets.

and contributed to testing alongside Pietro Signorino. Paolo Cursi was responsible for generating the LUPerson dataset and designing the custom ViTvae architecture, while Michele Palma led

the pretraining of the LUPerson model. Pietro Signorino contributed to testing, generated the CUHK03 dataset and solved the self-matching issue with the market-1501 dataset.